ARTIFICAL INTELLIGENCE PORTFOLIO

POKER VISION



INTRODUCTION

Welcome to the documentation for Poker Vision, a cutting-edge React Native application integrating live computer vision, YOLOv5 object detection, and real-time win probability calculations for a dynamic Texas Hold'em poker experience. This app excels in ultra-fast card and player detection, providing instantaneous insights into win probabilities during gameplay.

Main CHALLENGES

- **Ultra-Fast Detection:** Utilizes YOLOv5 for sub-second detection of playing cards and players, ensuring a responsive user experience.
- **Real-Time Win Probability:** Advanced algorithms calculate and display real-time win probabilities, adding strategic depth to the poker experience.
- **Cross-Platform Compatibility:** Built with React Native, the app offers a seamless experience on both Android and iOS platforms.

MAIN FEATURES

- **Ultra-Fast Detection:** Utilizes YOLOv5 for sub-second detection of playing cards and players, ensuring a responsive user experience.
- **Real-Time Win Probability:** Advanced algorithms calculate and display real-time win probabilities, adding strategic depth to the poker experience.
- **Cross-Platform Compatibility:** Built with React Native, the app offers a seamless experience on both Android and iOS platforms.



SOLUTION

- **Player Recognition Algorithms:** Implemented advanced algorithms for accurate real-time player recognition and tracking.
- **Enhanced Card Detection Models:** Fine-tuned YOLOv5 models to improve precision in detecting and identifying individual playing cards.
- **Parallel Processing and Optimization:** Leveraged parallel processing techniques and optimized algorithms for real-time win probability calculations.

SERVICES

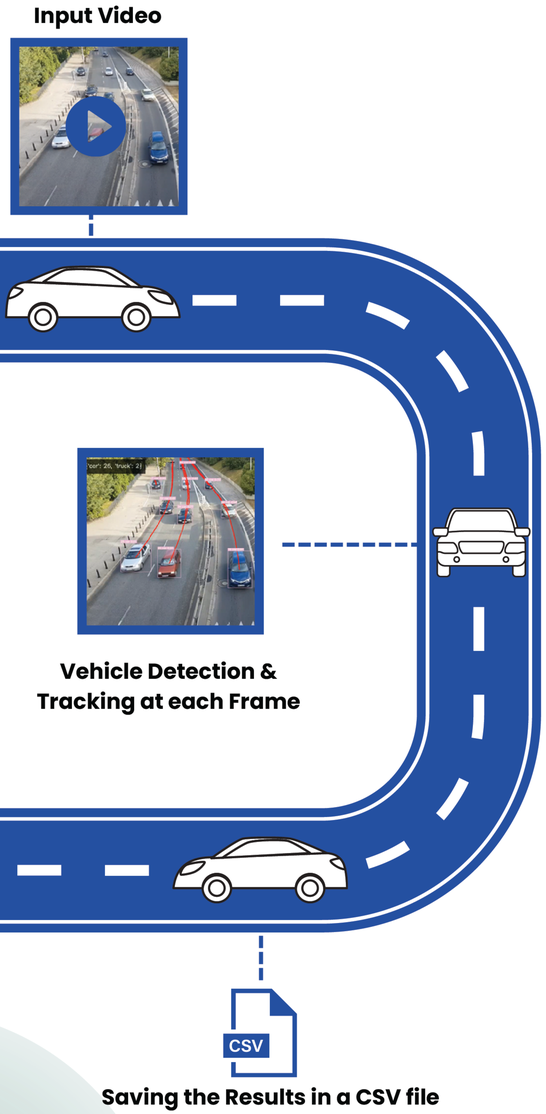
- **Ultra-Fast Object Detection:** Cutting-edge solutions for sub-second card and player detection.
- **Real-Time Win Probability Calculations:** Strategic insights through efficient algorithms.
- **Custom Development and Consulting:** Tailored services and consultation for similar computer vision projects.

Business Benefits

- **Immersive Gaming Experience:** Accurate player and card detection contribute to an immersive poker experience.
- **Real-Time Responsiveness:** Sub-second detection and dynamic win probabilities enhance gameplay responsiveness.
- **Cross-Platform Reach:** The app seamlessly runs on Android and iOS, expanding user reach.

TECHNOLOGIES

PyTorch roboflow python YOLO iOS android Java



V-TRACK

INTRODUCTION

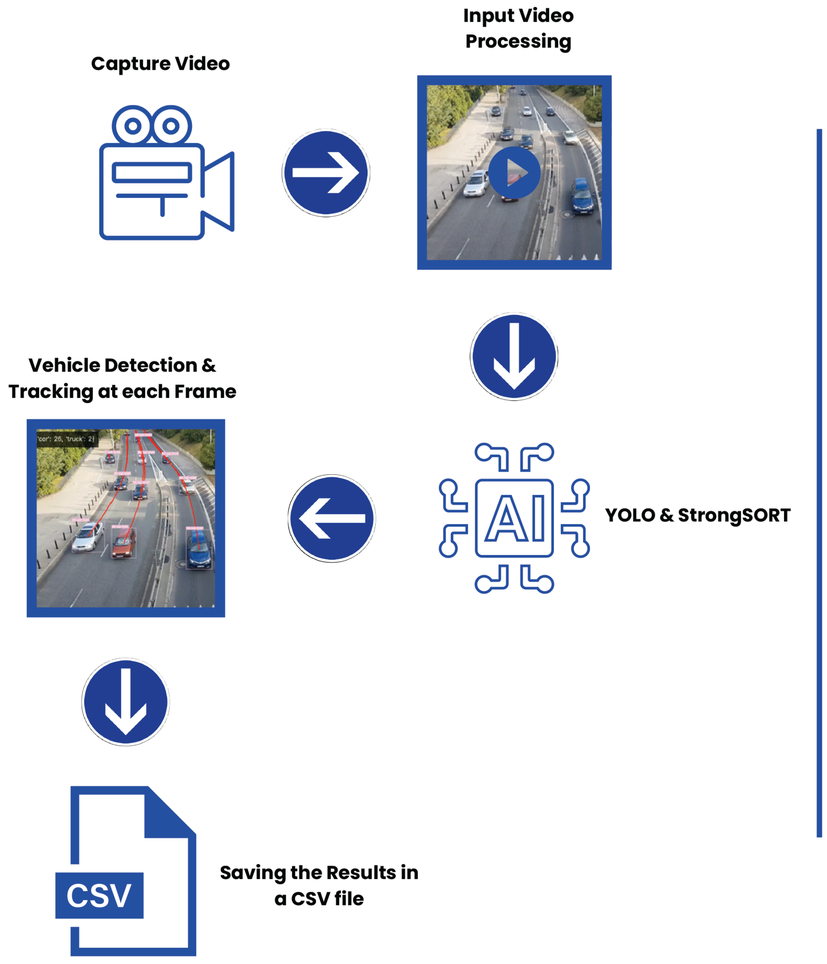
Experience the future of intersection traffic monitoring with V Track - the intelligent and AI-powered solution that keeps you on the right path.

MAIN CHALLENGES

- Traffic congestion is a major problem in cities worldwide, causing significant economic and environmental costs.
- Intersections are often high-risk areas for accidents, and vehicle tracking technology can help identify potentially dangerous situations before they occur.
- Decision-making in dynamic disaster situations.

SOLUTION

- A deep learning based efficient solution for monitoring traffic flow at intersections using AI-powered vehicle tracking. By detecting and tracking vehicles in real-time, we can optimize traffic flow and provide real-time traffic updates to drivers, leading to reduced congestion, improved safety, and saved time.
- By optimizing traffic flow at intersections using AI-powered vehicle tracking, our project can help reduce congestion and alleviate the negative impact it has on people's daily lives.
- AI-based decision support systems can analyze data from various sources to provide real-time insights and recommendations to disaster response teams, facilitating effective decision-making in dynamic disaster situations.
- By alerting drivers and city officials of potential safety hazards, our project can help prevent accidents and improve overall road safety.



SERVICES

Enhanced traffic flow optimization, Real-time accident detection and response, Traffic violation monitoring and enforcement, Traffic pattern analysis and planning, Intelligent transportation systems integration, Data-driven decision-making, Improved pedestrian safety, Traffic congestion reduction, Traffic monitoring for law enforcement, Environmental impact analysis

BUSINESS BENEFITS

- Efficient traffic incident response through real-time monitoring and data analysis.
- Accurate accident reconstruction and prevention using vehicle tracking data.
- Data-driven traffic management strategies for improved safety and efficiency.

TECHNOLOGIES

StrongSORT

UMRAH CONNECT



INTRODUCTION

Umrah Connect is a dedicated platform designed to assist and support pilgrims embarking on the sacred Umrah journey. With a primary focus on providing authentic and reliable information, Umrah Connect serves as a communication bridge for pilgrims, offering guidance and support throughout the entire Umrah process. Whether it's planning ahead or seeking assistance during the pilgrimage, Umrah Connect is there every step of the way.

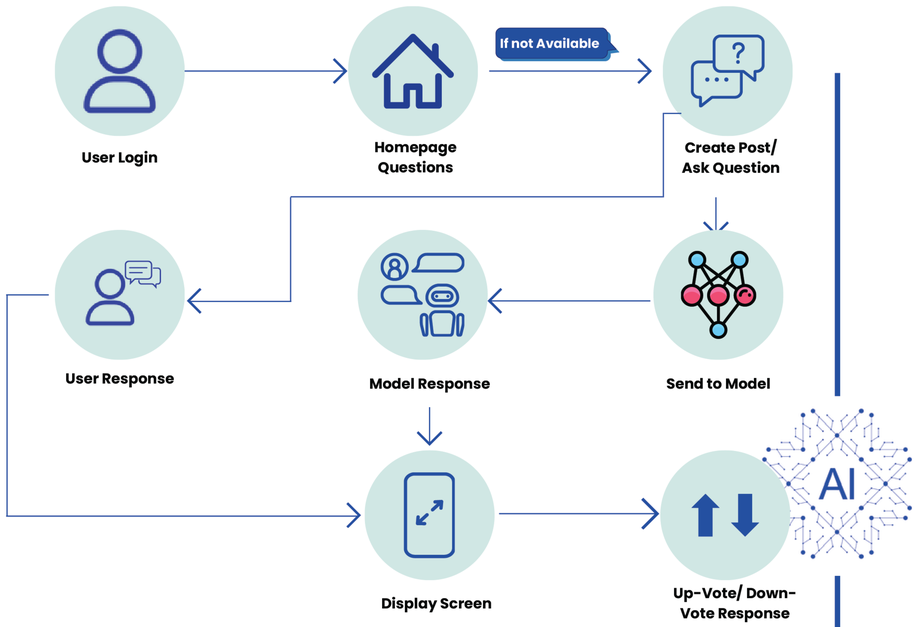
MAIN CHALLENGES

- Time-consuming and daunting nature of the Umrah pilgrimage process
- Difficulty in finding accurate information and distinguishing its authenticity

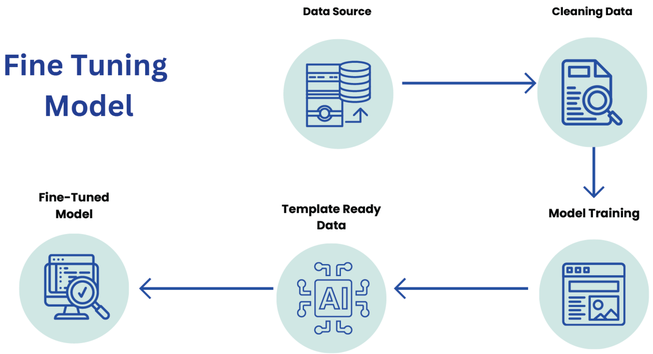
SERVICES

- Access to authentic and verified information in real-time
- Utilization of advanced LLM model for query responses
- Integration of speech-to-text functionality for seamless query submission
- Android app developed in Kotlin for user convenience
- Deployment on the Google Play Console
- Google Sign-In for secure user authentication

App flow



Fine Tuning Model



BUSINESS BENEFITS

- Reliable access to authentic information throughout the Umrah journey
- Time-saving solutions for pilgrims by providing quick responses to queries
- Enhanced user engagement through community interaction and feedback mechanisms

SOLUTION

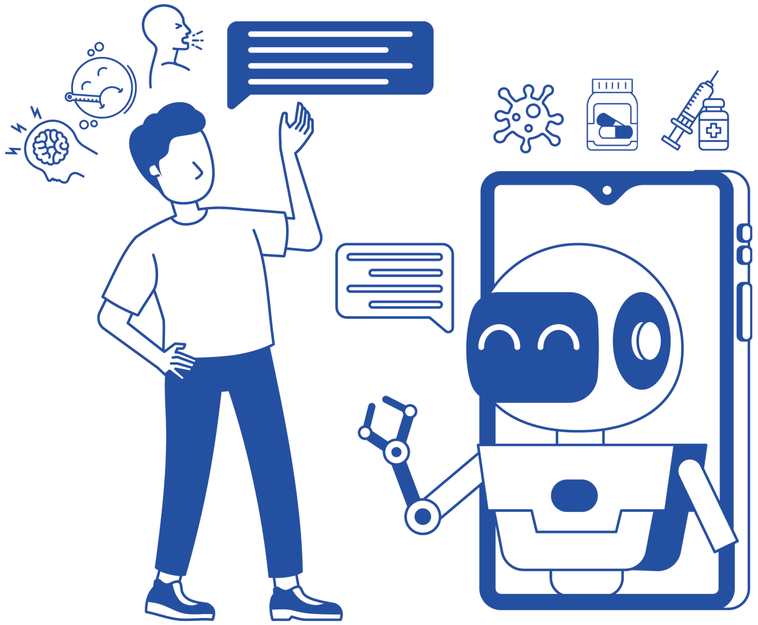
- User-friendly mobile app where pilgrims can submit queries
- Advanced LLM (Large Language Model) system trained on authentic datasets
- Community-driven approach with engagement features like comments, upvotes, and downvotes

TECHNOLOGIES

Kotlin · REST API · OpenAI · Google Play Console

HealthQuery

Revolutionizing Patient-Doctor Interactions



INTRODUCTION

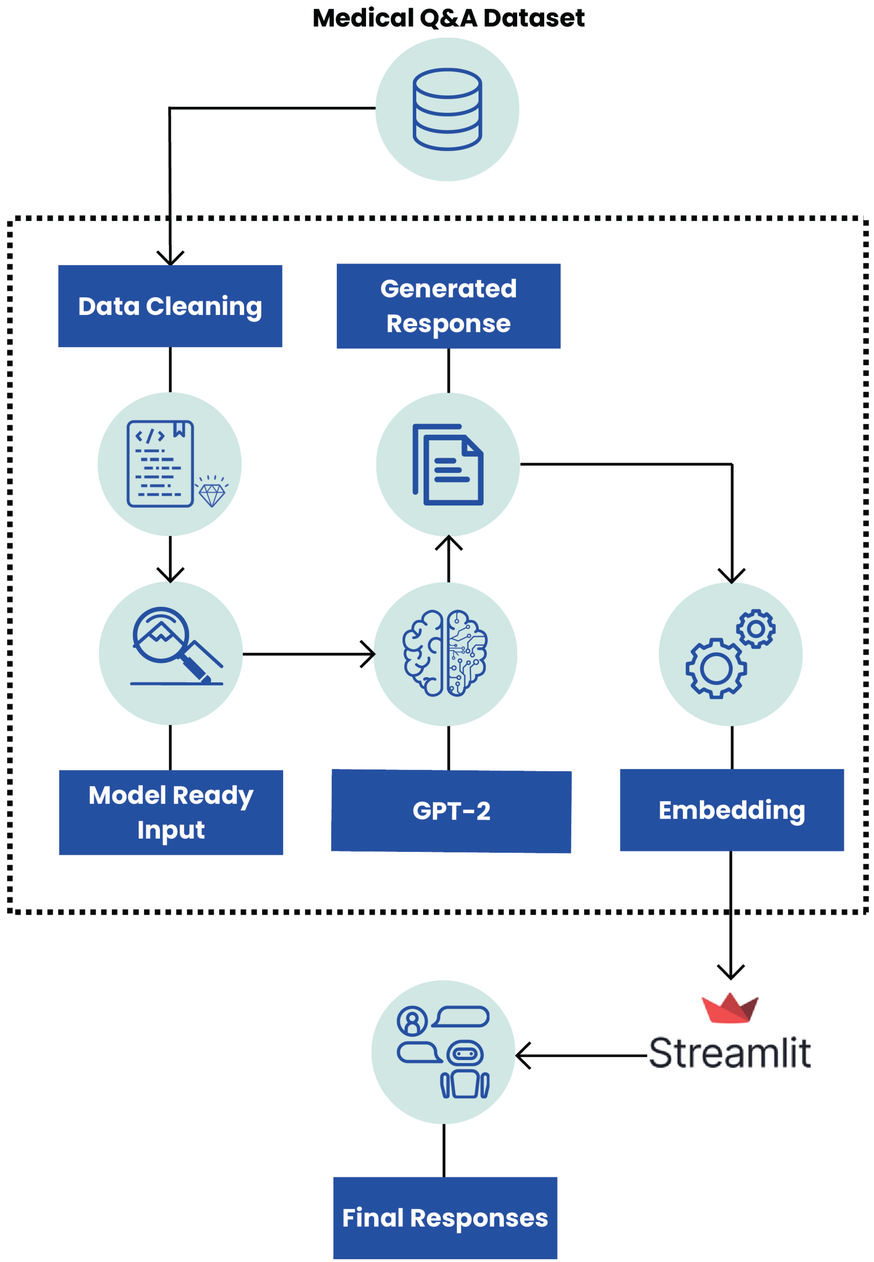
HealthQuery is an AI-driven medical question and answer (Q&A) system that empowers patients to seek precise medical information and advice without requiring immediate doctor consultation. The system is built using the GPT2 model, which has been trained on a medical conversational and descriptive dataset comprising 0.2 million rows of question-answer pairs. It is deployed on Streamlit using Ngrok, providing a user-friendly interface for seamless interaction.

MAIN CHALLENGES

- Long wait times for doctor appointments.
- Limited access to medical expertise.
- Difficulties in obtaining timely and accurate information about health concerns.

SOLUTION

- AI-driven medical Q&A system.
- Accurate and accessible knowledge.
- Generates prompt and reliable responses to medical queries.



HealthQuery

Revolutionizing Patient-Doctor Interactions

SERVICES

- Input medical questions and receive relevant and accurate responses.
- GPT-2 generates contextually appropriate and informative answers.
- User-friendly web application interface.

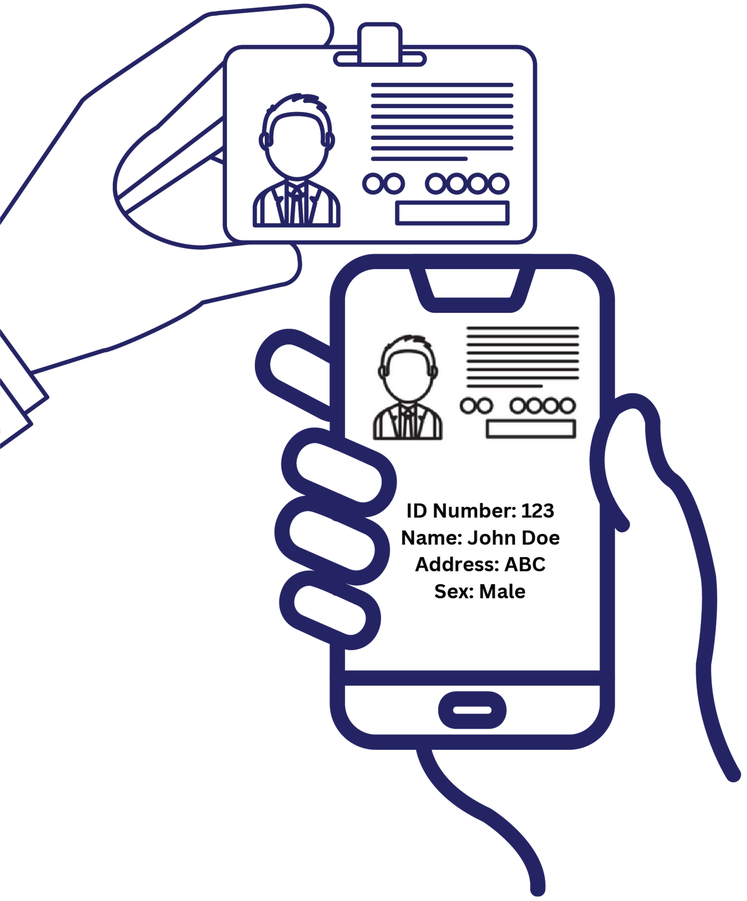
BUSINESS BENEFITS

- Reduces burden on healthcare professionals.
- Improves doctor-patient interactions.
- Optimizes healthcare resource allocation.
- Saves time and resources for patients and providers.

TECHNOLOGIES

PyTorch GPT-2 Streamlit ngrok Hugging Face kaggle

ACCUSCAN



INTRODUCTION

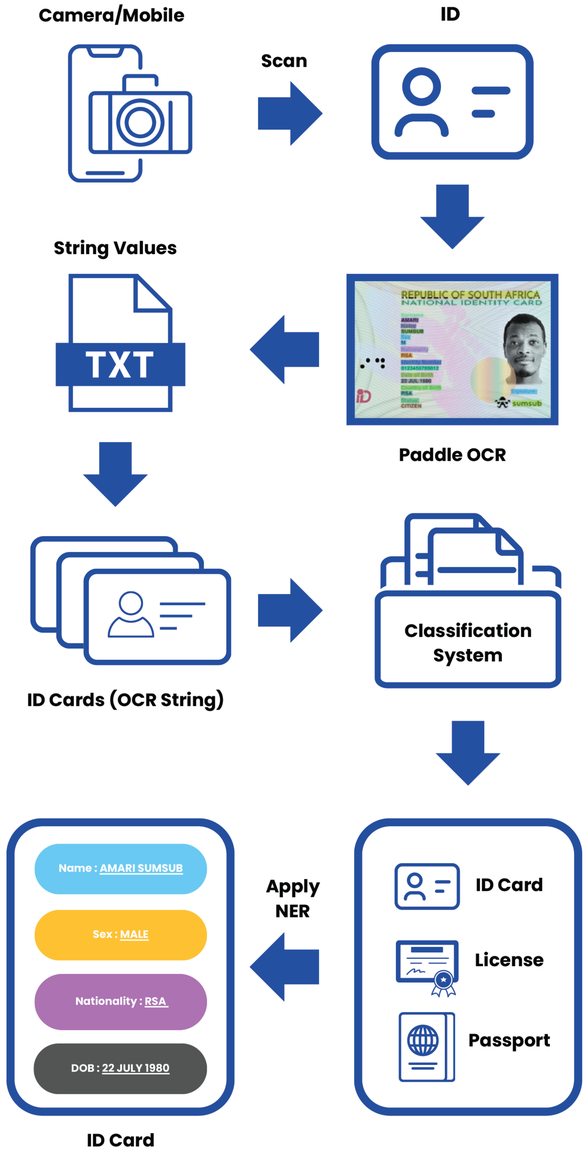
AccuScan represents an innovative project aimed at refining the data collection process by harnessing document scanning and automation technologies. By sequentially deploying advanced OCR, Text Classification, and NER algorithms, AccuScan effectively addresses the typical impediments of manual data entry, introducing streamlined solutions to enhance time efficiency, data accuracy, and overall workflow optimization.

MAIN CHALLENGES

- Manual data collection from documents is time-consuming due to the need for manual data entry.
- Collecting data manually introduces a higher risk of human errors and inconsistencies in the data.
- The process of manual data collection from documents involves multiple steps and can be inefficient, requiring physical document handling and storage.

SOLUTION

AccuScan transforms data extraction from scanned documents by automating manual entry, utilizing advanced OCR and Text Classification to swiftly extract information. Following OCR's accurate text extraction, the system employs Text Classification to identify the scanned card type, triggering a specialized Named Entity Recognition (NER) for precise entity extraction. This process enhances accuracy, expediting data collection.



AccuScan's services encompass document scanning, OCR text extraction, card-specific Text Classification, tailored NER, data validation, seamless system integration, and ongoing support, resulting in efficient and error-minimized data management.

SERVICES

AccuScan offers the following services:

- Document scanning for identity documents.
- OCR extraction of text from scanned documents.
- Text Classification based on extracted String
- Named Entity Recognition (NER) for entity extraction.
- Data validation and verification.
- Data integration with existing systems.
- Customization options and ongoing support.

BUSINESS BENEFITS

- By automating data collection and eliminating manual entry, businesses can streamline processes and save time.
- Utilizing advanced technologies, data accuracy is enhanced, minimizing errors and providing reliable information.
- The digital approach of AccuScan simplifies workflows, promotes collaboration, and optimizes overall operations.

TECHNOLOGIES

spaCy (PaddleOCR), **FastDeploy**

FaceVerifyX



INTRODUCTION

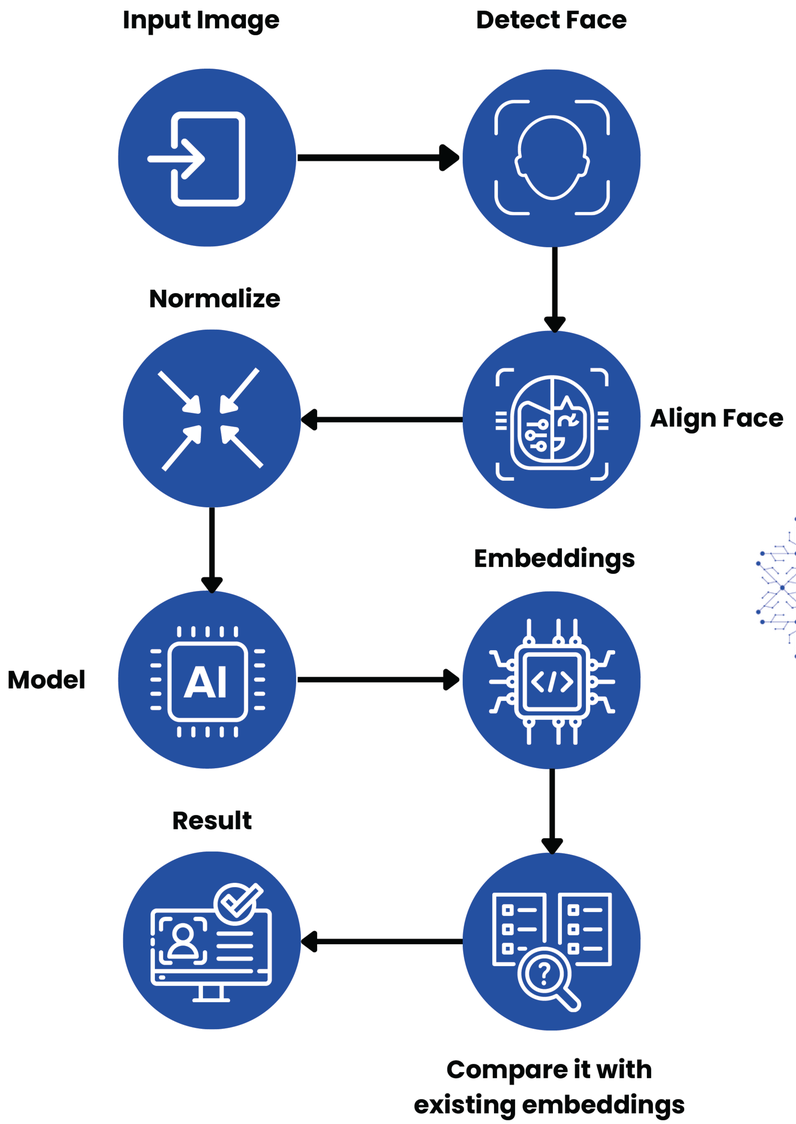
FaceVerifyX offers a dependable and secure method to confirm your identity, protecting your information from the vulnerabilities of easily compromised passwords. Our mission is to provide you with a straightforward, secure, and seamless Face Recognition service that's accessible to all while ensuring the utmost safety and integrity of your identity.

MAIN CHALLENGES

- Accurately recognizing the faces of individuals with darker skin tones poses a significant challenge, as current methods often display biases that can result in inaccuracies.
- Traditional face verification methods can be time-consuming, causing delays in critical scenarios.

SOLUTION

- We've meticulously designed FaceVerifyX to combat racial biases, ensuring unbiased and consistent results to foster inclusivity and fairness.
- Our real-time face recognition capabilities address the need for swift decision-making, empowering organizations with rapid and reliable verification solutions.



SERVICES

Our versatile FaceVerifyX finds applications in various use cases, including:

- *Employee Attendance Tracking*: Efficiently track attendance while minimizing fraud.
- *Smartphone Unlocking*: Enhance security with seamless face-based unlocking.
- *Digital Identity Verification*: Offer secure and convenient digital identity verification.
- *Financial Transactions*: Strengthen financial transaction security.
- *User Onboarding*: Streamline onboarding processes with swift and reliable face authentication.
- *Remote Verification*: Enable trustworthy remote identity verification

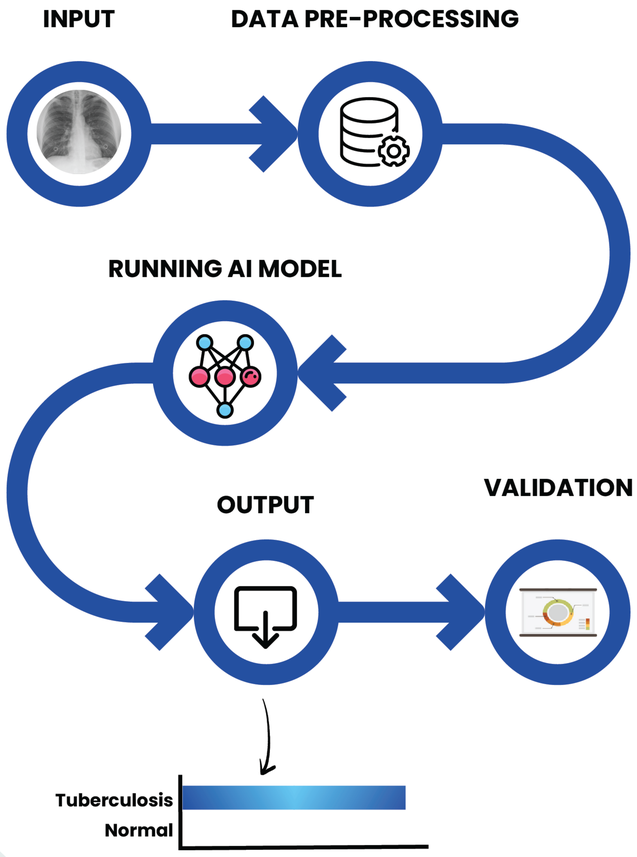
Business Benefits

FaceVerifyX provides a robust face recognition solution by leveraging cutting-edge technology, saving time and resources, ensuring fairness, enhancing security, and offering customization for specific client needs.

TECHNOLOGIES

deepFace

TUBERCO-LENS



INTRODUCTION

Revolutionary AI-Technology for efficient diagnosis of TB through X-Ray Images. A deep learning model to identify Tuberculosis cases from chest x-rays. The model is able to classify Tuberculosis X-ray images from normal chest X-ray images.

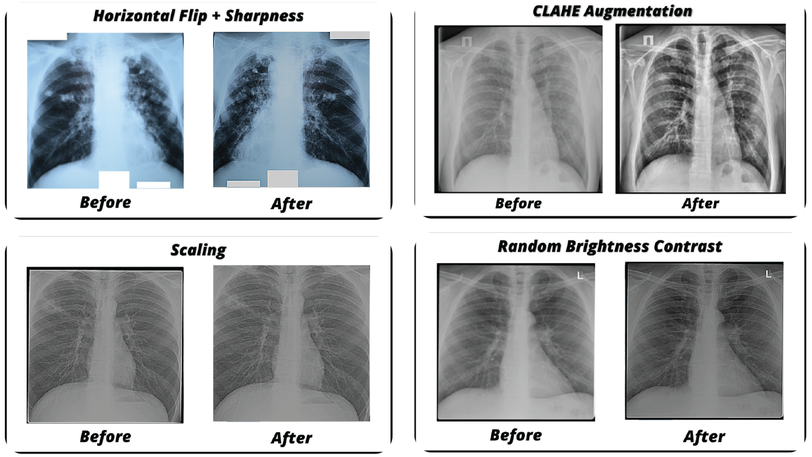
MAIN CHALLENGES

- In many parts of the world, diagnosing TB can be challenging and time-consuming, particularly in areas with limited resources or a high TB burden
- The dataset used in this study was imbalanced in terms of images.
- There were **3500 Images** of Normal Chest X-Ray and **700 images** of TB effected X-Rays.

SOLUTION

- Step 1: As input X-Ray images of the effected lungs are collected.
- Step 2: Dataset was balanced using the image augmentation and focal loss.
- Step 3: AI Deep Learning Model was used to process images.
- Step 4: The output showed whether the lungs is effected with TB or not.

HOW THE DATA WAS BALANCED?



SERVICES

X Ray Classification Solutions, Custom Model Development, Consulting and Advisory Services for similar projects, Model Evaluation and Performance Analysis

BUSINESS BENEFITS

- Enhanced healthcare decision-making: Accurate TB classification using our deep learning model supports informed treatment decisions, improving patient management and outcomes.
- Improved patient outcomes: Early detection with our deep learning model reduces TB spread, enhances treatment success, and decreases healthcare costs.
- Cost-effective healthcare delivery: By reducing misdiagnosis rates, our model enables prompt treatment initiation, leading to cost savings and optimized resource allocation.

TECHNOLOGIES

Neurosight





INTRODUCTION

Embark on a journey into the cutting-edge domain of medical technology with our brain tumor classification project, driven by the potent MobileNetV2 AI model. This endeavor represents a fusion of advanced artificial intelligence and medical diagnosis. The project stands as an intersection of innovation and healthcare, showcasing the potential of AI to revolutionize early detection and informed medical assessments.

MAIN CHALLENGES

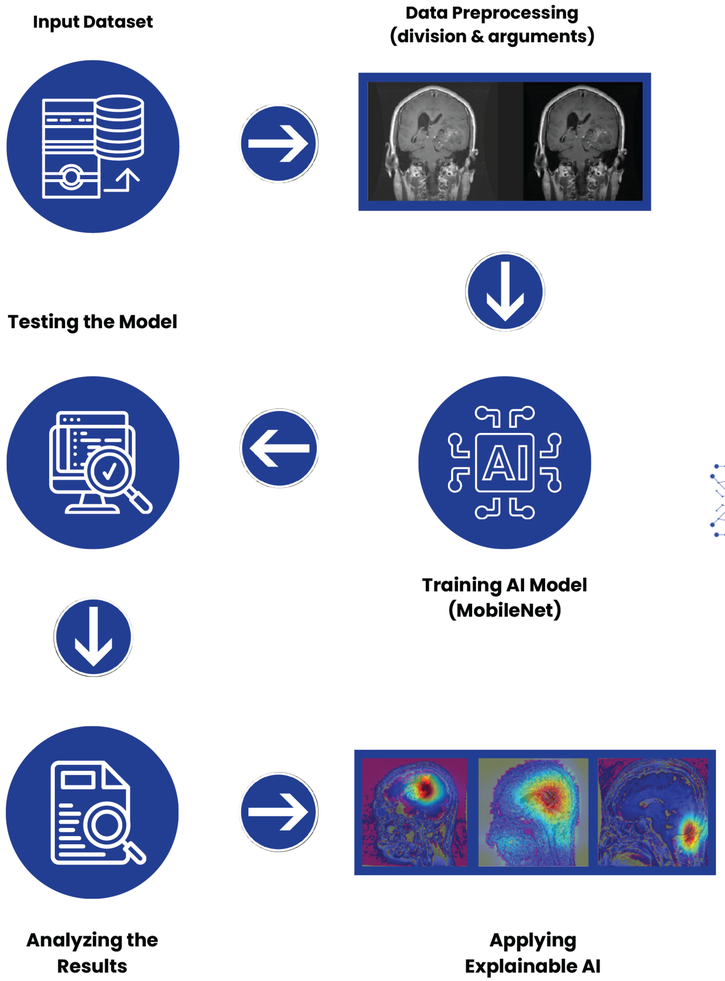




- Dataset used in this project has images from different angles. Some from top angle, some from back angle and some from left angle.
- Dataset was not balanced. Training and testing datasets have imbalanced number of images.
- Applying explainable AI to better understand model's working.

SOLUTION

- Following dataset acquisition, segmentation into subclasses, such as glioma_tumor_top_angle, and meningioma_tumors_left_angle and so was undertaken.
- This subdivision strategically addresses the challenge posed by variations in image angles.
- Augmentation techniques were subsequently employed to effectively enrich the dataset.
- This augmentation process significantly enhances data diversity, ultimately boosting the robustness of our approach.



- AI model – Mobilenet-V2 model was applied for classification.
- Grad-cam was used to apply explainable AI on model for better understanding of model working.

SERVICES

Accurate Brain Tumor Classification, Custom Data Segmentation, Data Augmentation Expertise, AI Model Integration, Explainable AI Insights, Consultation and Training, Ongoing Support.

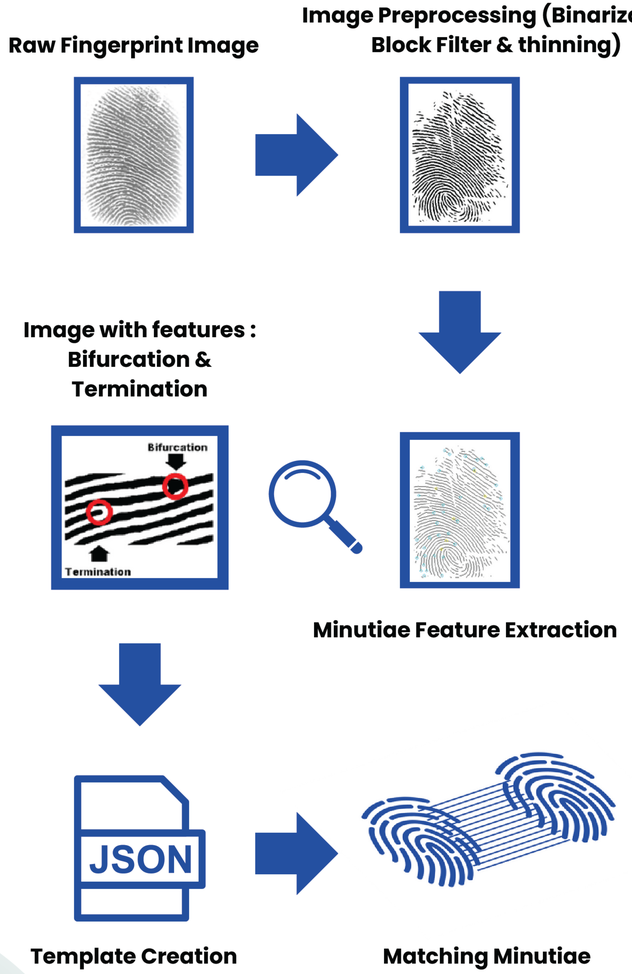
BUSINESS BENEFITS

- Brain tumor classification project significantly enhances the accuracy of diagnoses, reducing the potential for misdiagnoses and facilitating more informed medical decisions. This accuracy directly translates to improved patient care and outcomes.
- By automating the classification process using our AI model, medical professionals can save valuable time that would otherwise be spent manually analyzing brain scans.
- This AI model alleviates the burden on medical professionals by handling the intricate task of classifying brain tumors. This empowers healthcare teams to focus on personalized patient care and complex decision-making, enhancing the quality of healthcare services.
- By reducing the need for manual labor in image analysis, this project contributes to cost savings in the long run. Fewer resources are required for repetitive tasks, and the investment in AI technology yields returns through improved efficiency.

TECHNOLOGIES

GradCam

AuthenticaPro



INTRODUCTION

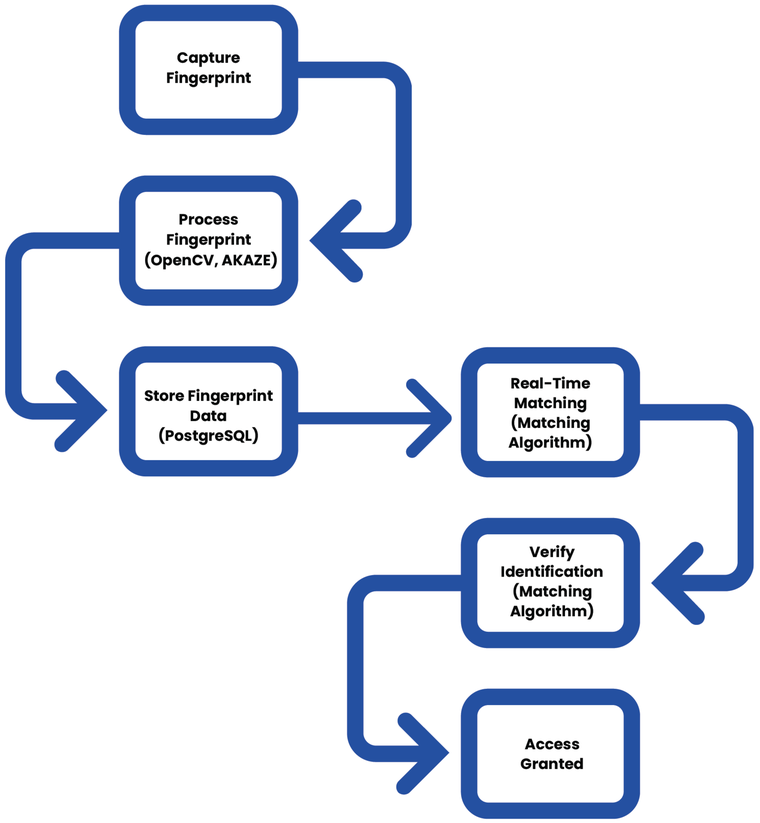
The Fingerprint Authentication System is a comprehensive solution designed to provide secure user identification and verification. It leverages advanced technologies such as OpenCV, the AKAZE algorithm, Django, Postgresql, Docker, nginx, and gunicorn to create a reliable fingerprint scanner. The system is accessible through an Android app that communicates with an AWS server using APIs.

MAIN CHALLENGES

- Capturing and processing high-quality fingerprint images.
- Developing an efficient matching algorithm for real-time identification.
- Ensuring secure storage and retrieval of fingerprint data.
- Integrating the system with an Android app and AWS server for seamless user access.

SOLUTION

- Utilizing OpenCV and the AKAZE algorithm for accurate fingerprint image capture and processing.
- Implementing a robust matching algorithm to perform real-time identification and verification.
- Utilizing Django and Postgresql to securely store and retrieve fingerprint data.
- Integrating the system with an Android app through API communication with an AWS server.



SERVICES

- Fingerprint capture and processing: The system captures high-quality fingerprint images using OpenCV and applies the AKAZE algorithm for feature extraction.
- Real-time matching: The system performs real-time matching and verification of captured fingerprints using a reliable matching algorithm.
- Secure storage: Fingerprint data is securely stored in a Postgresql database, ensuring confidentiality and integrity.
- API integration: The Android app communicates with the AWS server through APIs, enabling seamless access to the Fingerprint Authentication System.

BUSINESS BENEFITS

- Enhanced security: The system provides a robust and reliable method for user identification and verification, minimizing the risk of unauthorized access.
- Improved user experience: Users can authenticate themselves using their fingerprints, eliminating the need for traditional passwords.
- Time and cost savings: The automated fingerprint authentication process reduces the time and resources required for manual identity verification.
- Scalability: The system can handle a large number of users, making it suitable for organizations of various sizes.

TECHNOLOGIES

OpenCV docker gunicorn PostgreSQL NGINX aws

E-COMMERCEQUERY



INTRODUCTION

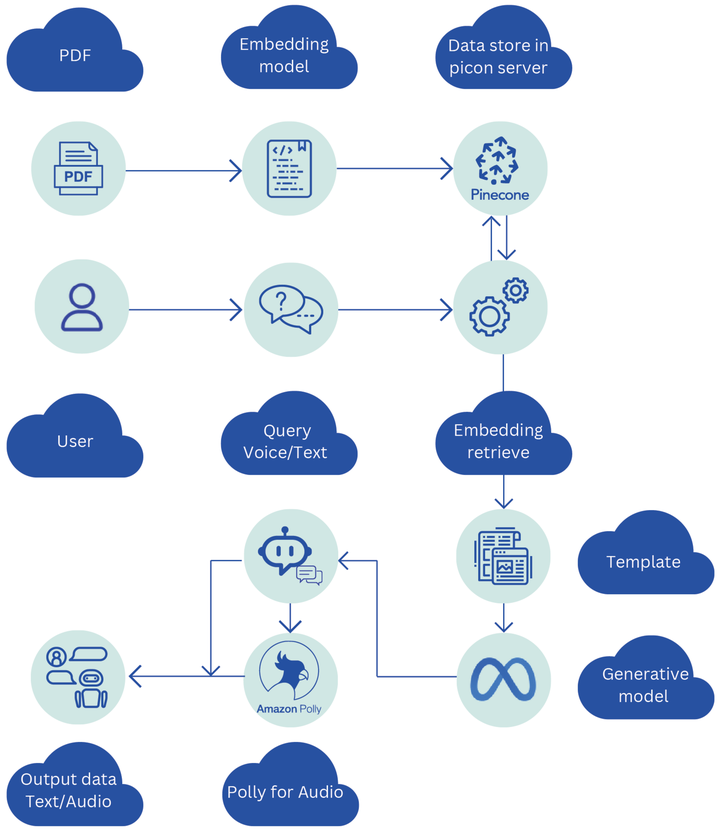
E-CommerceQuery is an innovative AI-driven Chatbot system designed to elevate customer support within the e-commerce sector. Utilizing state-of-the-art technologies such as LangChain, Llama 13b Quantize Model, Pinecone, Amazon Lex, Amazon Polly, Hugging Face, and AWS SageMaker, the system efficiently manages user queries. By extracting relevant information from an e-commerce details PDF through Hugging Face embedding model, it provides accurate responses in both text and audio formats.

MAIN CHALLENGES

- Inadequate customer support responsiveness
- Difficulty in retrieving specific e-commerce information
- Limited user engagement due to generic responses

SOLUTION

- AI-driven chatbot for e-commerce customer support
- Extraction of precise information from e-commerce PDFs using Hugging Face embedding model
- Context-aware responses for an enhanced user experience



SERVICES

- Query resolution with detailed e-commerce information
- LangChain for query and context storage
- Llama 13b Quantize Model for response generation
- Integration with Amazon Polly for text-to-speech
- Integration with Amazon Lex for Chabot functionality
- Seamless user interaction through text and audio responses

BUSINESS BENEFITS

- Enhanced customer satisfaction through accurate and timely responses
- Improved user engagement and loyalty
- Increased efficiency in handling customer queries
- Streamlined customer support processes



TECHNOLOGIES

LangChain · Amazon Lex · Amazon Polly · aws · Hugging Face · Amazon SageMaker · Pinecone

Llama 13b Quantize Model